

Optimal Scheduling of Data Center Virtual Power Plant in Electricity–Carbon Joint Market Under Uncertainty

Ziwei Zhao, Chen Yang, Chen Liang, Dan Xu, Yilin Zhang, and Junjie Tang, *Member, IEEE*

Abstract—As an emerging demand response (DR) resource, data centers (DCs) have garnered significant attention due to their ability to adapt to stochastic renewable energy fluctuations. To fully exploit the price signals from diverse markets and achieve the flexible complementarity between DCs and virtual power plant (VPP) resources, an optimal scheduling strategy is proposed herein for the data center virtual power plant (DCVPP) participating in the electricity–carbon joint market under uncertainty. Specifically, a refined model of cooling system power consumption in DC is developed based on the second-order equivalent thermal parameter (ETP) theory, which enhances the accuracy of load modeling and maximizes the potential for DR. To address the penalty risks associated with forecast uncertainties in renewable energy generation and load demands, a risk-based reserve scheme is developed by integrating the Gaussian copula with two-sided superquantile theory. This scheme provides a coordinated mechanism for quantifying the correlations between forecast errors. Additionally, an optimal scheduling strategy is formulated for DCVPPs participating in the electricity–carbon joint market, which coordinates DC load flexibility with distributed energy resources to optimize economic performance and reduce carbon emissions. Case studies demonstrate that the proposed DC load model improves the root mean square error by 57.9 to 68.8%, and the risk-based reserve scheme reduces penalty costs by 24.4%. Furthermore, the integration of the carbon trading mechanism in the scheduling strategy results in a \$259.9 increase in economic benefits and an average reduction of 14% in carbon emissions.

Index Terms—Data center, virtual power plant, demand response, uncertainty, electricity–carbon joint market, optimal scheduling.

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I. INTRODUCTION

WITH the rapid advancement of information technology and the widespread adoption of Internet-based applications, the energy consumption of data centers (DCs) has attracted increasing attention [1]. Unlike conventional power loads, the DCs exhibit pronounced temporal and spatial flexibility, making them act as a novel class of large-capacity demand response (DR) resource when real-time response capabilities and the flexible scheduling characteristics are fully considered [2]. Meanwhile, the increasing integration of renewable energy sources (RESs) introduces substantial operational uncertainty, and the coupling of electricity and carbon markets further increases system complexity. These challenges highlight the urgent need for advanced scheduling strategies that can enhance renewable energy utilization, reduce carbon emissions, and fully exploit the flexibility of DCs in future low-carbon energy systems.

Accurately modeling DC power consumption is essential for characterizing and exploiting its load flexibility. The total DC load consists of three primary components: IT equipment, cooling system, and other auxiliary equipment. The IT equipment is responsible for processing user workloads, whereas the cooling system ensures effective heat dissipation to maintain the safe and reliable operation of the IT equipment. Typically, the power consumed by the IT equipment and cooling equipment accounts for 80-90% of the total DC power consumption [3], [4]. Existing research on DC load modeling primarily focuses on the impact of IT equipment operating conditions on overall DC energy consumption. The prevailing modeling approaches include additive models [5], regression models [6], and linear models based on Power Usage Effectiveness (PUE) [7]. To accurately characterize the strong coupling between the power consumption of the IT equipment and the cooling system, the effects of ambient temperature variations on DC load behavior and DR capability must be explicitly taken into account. [8] proposes a DC load model based on the first-order equivalent thermal parameter (ETP) theory, but it fails to account for the indoor thermal mass differentiation, resulting in deviations between the simulated and actual load profiles under the dynamic operational conditions. Therefore, accurate DC load modeling requires an in-depth consideration of thermal environmental factors, which is essential for capturing load dynamics and

flexibility.

Unlike other flexible loads, DCs possess significant temporal load regulation potentials through batch workload scheduling mechanisms. Consequently, DC participation in DR programs has attracted increasing research interest, as it can enhance power system operation cost effectively and efficiently [9], [10]. [11] proposes that the DCs could minimize the carbon footprint and power cost by delaying the processing of flexible workloads. [12] utilizes the scheduling flexibility of workloads and proposes a method for the DC workload scheduling, which effectively reduces the DC operating cost. Recently, as an effective solution for integrating distributed energy resources, data center virtual power plants (DCVPPs) have attracted rising attention in the DC energy management [13]. [14] proposes an optimal power dispatch problem with a multisource virtual power plant (VPP) containing DC for maximizing the revenue. Considering the operational flexibility of the workloads and power supply devices, [15] builds a scheduling framework for DC-based VPPs. However, fully exploiting the operational flexibility of DCs and achieving coordinated optimization with internal flexible resources in DCVPPs to meet low-carbon economic objectives remains an open research challenge.

Participation in electricity markets under uncertainties arising from RESs and load forecast errors introduces significant risks to the decision-making strategy of DCVPP. To hedge against the risks, the theory of quantile and superquantile, also known as the value at risk (VaR) and the conditional value at risk (CVaR), is employed to quantify the required reserve capacity and the expected value exceeding the reserve capacity [16]. In [17], the CVaR is employed in the stochastic model to manage the energy and reserve capacity, as well as to control the risk of VPP profit variability. By adopting the superquantile theory, [18] characterizes the economic penalties beyond the reserve capacity and optimizes the allocation of reserve capacity. It is found that the uncertainty is tackled by a confidence interval subjectively in the existing research, and therefore, the risks caused by that beyond the confidence interval and the resulting economic penalties are not well quantified [19]. The temporal variability of penalties for the power deviations in electricity markets requires the dynamic optimization of the upward and downward reserve capacity of such DCVPP. Furthermore, the forecasting error parameters of different types of RESs and loads are mostly set independently in the existing studies, and the influence of the correlation between RES and load forecast errors on the reserve capacity is neglected [20]. Therefore, it is necessary to explicitly account for the correlation among forecast errors of RESs and loads in uncertainty modeling, to investigate and analyze their interactive impacts on the day-ahead scheduling of DCVPP. The comparisons with existing literature are summarized in Supplementary Material Table S- I .

Accordingly, research on the optimal scheduling of DCVPP holds significant potential for effectively integrating the flexible regulation capabilities of DC loads with other distributed energy resources. Meanwhile, the evolution of electricity and carbon markets has led to increasingly com-

plex couplings among the energy, ancillary service, and carbon markets. This trend positions DCVPP as a pivotal participant in future electricity-carbon joint market transactions. To address the aforementioned research gaps, this paper proposes an optimal scheduling strategy for the DCVPP participating in joint electricity-carbon markets under uncertainty. As shown in Fig. 1, the main contributions are summarized as follows:

1) To evaluate the DR potential of DC, a refined model of cooling system power consumption in DC is established based on the theory of second-order ETP. This model improves the accuracy of DC modeling, thus enabling more effective exploitation of load regulation potential and enhancing the operational flexibility of the DCVPP.

2) To mitigate penalty risks arising from forecast uncertainties in RESs and loads, a risk-based reserve scheme is developed for DCVPP participation in the electricity market. By integrating the Gaussian copula with the two-sided superquantile theory, the scheme quantifies risks associated with both excess and insufficient power while accounting for forecast error correlations.

3) An optimal scheduling strategy is proposed for the DCVPP participating in the electricity-carbon joint markets. By coordinating the DR capability of DC load with the operational flexibility of distributed resources, the proposed strategy maximizes the DCVPP revenue in the electricity-carbon joint market while reducing carbon emissions.

The remainder of this article is structured as follows. Section II presents the operational framework of DCVPP and the electricity-carbon joint market trading mechanisms. The modeling of DC is built in Section III. Section IV introduces the optimal scheduling strategy for the DCVPP participating in the electricity-carbon joint market. Section V focuses on the case studies. The conclusion is drawn in Section VI.

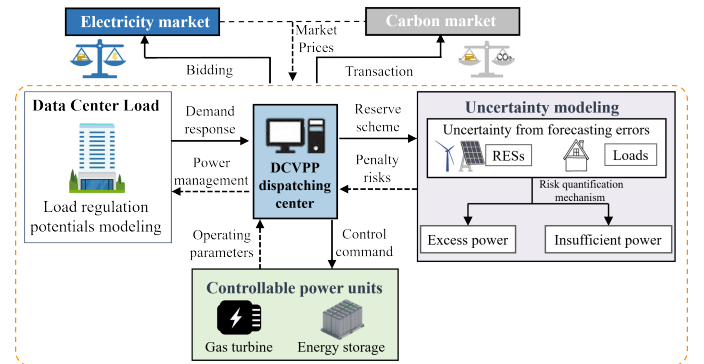


Fig. 1. Power management framework of DCVPP.

II. OPERATIONAL FRAMEWORK OF DCVPP AND ELECTRICITY-CARBON JOINT MARKET TRADING MECHANISMS

A. Operational Framework of DCVPP

This section presents the operational framework of a DCVPP in the electricity - carbon joint market. The DCVPP aggregator is assumed to manage a diversified portfolio of resources within a defined region, including DC, gas turbine (GT), wind turbine (WT), photovoltaic generator (PV), ener-

gy storage (ES), and rigid load (RL). As illustrated in Fig. 2, the DCVPP realizes the dispatching operation of energy equipment and implementation of the DR program at the DC side through the centralized control dispatching center, thereby meeting the power demand of all users within the system.

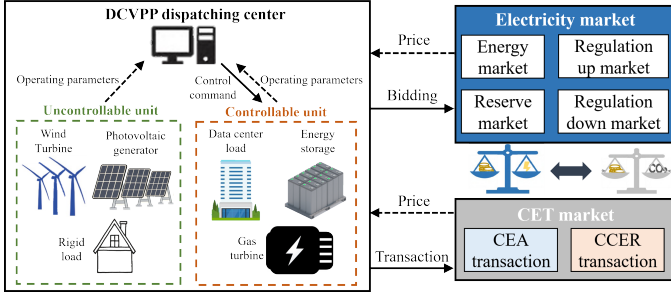


Fig. 2. Operational framework of DCVPP in electricity-carbon joint market.

Furthermore, the DCVPP is modeled as a price-taker player during the decision-making process of day-ahead optimal energy management [21]. Based on day-ahead forecasts, the DCVPP can purchase or sell electricity in the day-ahead electricity market (DAM) and provide ancillary services to obtain corresponding revenues. Meanwhile, it can also participate in the carbon emissions trading (CET) market to promote renewable energy consumption. Considering the coupling between the electricity market and the CET market, it is assumed that the period in the cycle of carbon allowances and carbon clearance is assumed to be one day, which is aligned with the time scale of the DAM [22].

B. Day-ahead Electricity Market

By participating in the DAM as a flexible power plant, the DCVPP can benefit from strategically selling the electricity or providing auxiliary services in response to varying market prices. In the electricity-carbon joint market framework considered in this paper, the options of electricity market participation are based on those available in the Texas power market from the United States, including the energy, reserve, and regulation (both up and down) markets [18].

C. Carbon Emissions Trading Market

The CET mechanism treats the carbon emission rights as tradable commodities among emitters. When actual carbon emissions exceed the allocated allowances, emitters must either purchase additional allowances or incur penalty charges. In the CET market, the two primary trading products are Carbon Emission Allowances (CEAs) and China Certified Emission Reductions (CCERs) [23].

CEA trading begins with the regulator setting a cap on the total CEAs in the market. These allowances are initially allocated to individual emission sources, which may subsequently trade the allowances based on the prevailing CEA price [24]. The CEAs of the DCVPP are calculated as:

$$Q_t^{\text{CEA}} = \eta \cdot P_t^{\text{GT}} \quad (1)$$

where Q_t^{CEA} is the total amount of CEAs for the GT in DCVPP, η is the CEA factor per unit of electricity for the GT, and P_t^{GT} is the generating power of GT at time t .

The CCERs obtained from RESs in the DCVPP can be traded in the CET market, as calculated in what follows [25]:

$$Q_t^{\text{CC}} = \sum_{R \in \{\text{WT, PV}\}} (\varepsilon^{\text{OM}} \cdot \phi_R^{\text{OM}} + \varepsilon^{\text{BM}} \cdot \phi_R^{\text{BM}}) P_t^R \quad (2)$$

where Q_t^{CC} is the obtained CCERs of DCVPP; ε^{OM} and ε^{BM} are the weights of the operating margin and build margin emission factors, respectively; ϕ_R^{OM} and ϕ_R^{BM} denote the operating margin and build margin emission factors, respectively; P_t^R is the generating power of RES R at time t .

Carbon emissions within the DCVPP can be calculated as:

$$Q_t^{\text{CE}} = \chi P_t^{\text{GT}} + \kappa (P_t^{\text{chpur}} + P_t^{\text{Lpur}}) \quad (3)$$

where Q_t^{CE} is the actual carbon emission of DCVPP, χ is the carbon emission factor of GT; κ is the carbon emission factor of the grid thermal power unit; P_t^{chpur} and P_t^{Lpur} are the ES's charging power and the loads' power through purchasing from the retail market, respectively.

III. DATA CENTER MODELING

A. Workload Model

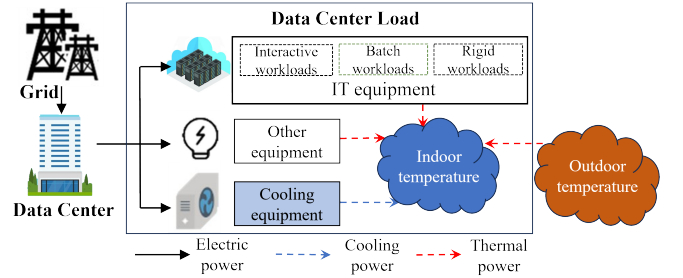


Fig. 3. Practical architecture of DC operation.

Fig. 3 presents the practical architecture of DC operation. Workloads arriving in the DC can be classified into three categories: interactive, batch, and rigid workloads [16]. Interactive workloads require immediate processing. Batch workloads allow for flexible service times and can be delayed by hours, whereas the rigid workloads impose stringent requirements on either processing time or quality of service.

$$E_t^{\text{org}} = \lambda_t^{\text{org}} + \omega_t^{\text{org}} + \zeta_t^{\text{org}} \quad (4)$$

where E_t^{org} , λ_t^{org} , ω_t^{org} , and ζ_t^{org} represent the original amounts of total workloads, interactive workloads, batch workloads, and rigid workloads arriving in DC at time t , respectively.

Batch workloads can be scheduled for execution within a specified time window, and hence their temporal shifting characteristics can be described as:

$$\begin{cases} \omega_t^{\text{flex}} = \omega_t^{\text{org}} + \sum_{t'=1}^{t-1} \omega_{t'} - \sum_{t''=t+1}^T \omega_{t''} \\ \sum_{t''=t+1}^T \omega_{i,t''} \leq \omega_{i,t}^{\text{org}} \end{cases} \quad (5)$$

where $\omega_{i,t}$ denotes the batch workload migrated from time t' to t ; $\omega_{i,t''}$ represents the batch workload transferred out from time t to t'' ; and ω_t^{flex} is the amount of batch workload processed at time t after flexible shifting.

B. Modeling of DC Load Based on Second-order ETP Theory

The total power demand of DC at time t , denoted as P_t^{DC} , can be decomposed into three components: the power consumption of IT equipment P_t^{IT} , the power consumption of cooling system P_t^{cool} , and the power consumption of other auxiliary equipment P_t^{others} , which can be expressed as:

$$P_t^{\text{DC}} = P_t^{\text{IT}} + P_t^{\text{cool}} + P_t^{\text{others}} \quad (6)$$

The power consumption of IT equipment is determined by the number of active servers and the corresponding workloads, with all servers assumed identical and the workloads evenly distributed. Thus, the power consumption of IT equipment is given by [26]:

$$P_t^{\text{IT}} = n_t^{\text{on}} P_{\text{idle}} + (P_{\text{peak}} - P_{\text{idle}}) u_t n_t^{\text{on}} \quad (7)$$

$$u_t = E_t / (n_t^{\text{on}} \mu) \quad (8)$$

where n_t^{on} is the number of active servers; E_t denotes the actual processed workload at time t ; u_t is the CPU utilization rate of active servers; μ , P_{idle} and P_{peak} are the average service rate, idle power, and peak power of each active server, respectively.

To maintain the internal ambient temperature of DC, the cooling system supplies thermal cooling power to eliminate the heat generated by IT and other auxiliary equipment, as well as the heat transferred from the outdoor environment [8]. Based on the second-order ETP theory [27], the relations among indoor and outdoor temperatures, thermal cooling power, and the power consumption of IT equipment and other auxiliary equipment are modeled as:

$$\begin{cases} \frac{d\text{Tem}_t^a}{dt} = -\frac{R_a + R_s}{C_a R_a R_s} \text{Tem}_t^a + \frac{\text{Tem}_t^s}{C_a R_s} + \frac{\text{Tem}_t^o + R_a Q_t^{\text{DC}}}{C_a R_a} \\ \frac{d\text{Tem}_t^s}{dt} = (\text{Tem}_t^a - \text{Tem}_t^s) / (C_s R_s) \end{cases} \quad (9)$$

$$Q_t^{\text{DC}} = (k_t^{\text{IT}} P_t^{\text{IT}} + P_t^{\text{others}} - k^{\text{cop}} P_t^{\text{cool}}) \quad (10)$$

where Tem_t^a and Tem_t^s are the indoor air and solid temperature; Tem_t^o is the outdoor air temperature; Q_t^{DC} is the total thermal power of DC; k_t^{IT} is the heat factor of IT equipment; and k^{cop} is the coefficient of performance of the cooling system. The analytical recursive form of (9) can be expressed as [27]:

$$\begin{bmatrix} \text{Tem}_t^a \\ \text{Tem}_t^s \end{bmatrix} = \mathbf{A} \begin{bmatrix} \text{Tem}_{t-1}^a \\ \text{Tem}_{t-1}^s \end{bmatrix} + \mathbf{B} (\text{Tem}_t^o + Q_t^{\text{DC}} R_a) \quad (11)$$

$$\begin{cases} \mathbf{A} = \begin{bmatrix} M - \frac{b}{2} Z + (C_s R_s)^{-1} Z & (C_a R_s)^{-1} Z \\ (C_s R_s)^{-1} Z & M + \frac{b}{2} Z - (C_s R_s)^{-1} Z \end{bmatrix} \\ \mathbf{B} = \begin{bmatrix} 1 - M - \frac{b}{2} Z + (C_a R_a)^{-1} Z & 1 - M - \frac{b}{2} Z \end{bmatrix}^T \end{cases} \quad (12)$$

$$\begin{cases} b = (C_a R_a)^{-1} + (C_a R_s)^{-1} + (C_s R_s)^{-1} \\ c = (C_a C_s R_a R_s)^{-1} \\ Z = (e^{(-b + \sqrt{b^2 - 4c})\Delta t/2} - e^{(-b - \sqrt{b^2 - 4c})\Delta t/2}) / \sqrt{b^2 - 4c} \\ M = (e^{(-b + \sqrt{b^2 - 4c})\Delta t/2} + e^{(-b - \sqrt{b^2 - 4c})\Delta t/2}) / 2 \end{cases} \quad (13)$$

where $\mathbf{A}$ and $\mathbf{B}$ are parameter matrices in recursive form; C_a and C_s are the heat capacities of the indoor air and solid, respectively; R_a is the thermal resistance between the indoor air and the external environment, and R_s is the thermal resistance between the indoor solid and the indoor air.

Based on the combination of (10)-(13), the power consumption of cooling system is derived through the second-order ETP model as follows:

$$\begin{cases} Q_t^{\text{DC}} = \frac{(\text{Tem}_t^a - \mathbf{A}_{11} \text{Tem}_{t-1}^a - \mathbf{A}_{12} \text{Tem}_{t-1}^s) - \text{Tem}_t^o}{\mathbf{B}_{11} R_a} - \frac{\text{Tem}_t^o}{R_a} \\ P_t^{\text{cool}} = (k_t^{\text{IT}} P_t^{\text{IT}} + P_t^{\text{others}} - Q_t^{\text{DC}}) / k^{\text{cop}} \end{cases} \quad (14)$$

The power consumption of other auxiliary equipment typically constitutes only a small proportion of the total DC load. Therefore, it is assumed to be 10% of the combined power consumption of IT equipment and cooling systems.

IV. OPTIMAL SCHEDULING STRATEGY FOR DCVPP PARTICIPATING IN THE ELECTRICITY-CARBON JOINT MARKET

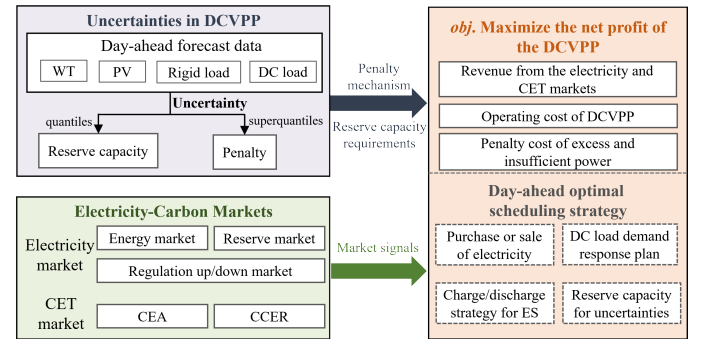


Fig. 4. Optimal scheduling strategy for DCVPP.

The overall structure of the proposed optimal scheduling strategy is illustrated in Fig. 4, which integrates uncertainty modeling, reserve allocation, and market participation into a unified framework. In practical DCVPP operations, the forecast errors in loads and RESs introduce substantial stochastic uncertainty into system behavior. To mitigate the resulting risks, a reserve capacity allocation mechanism utilizing controllable power resources (GT and ES) is adopted. Power deviations within the probabilistic confidence interval are fully compensated by the reserve capacity without incurring penalty costs. The penalty is evaluated through a Gaussian copula combined with the two-sided superquantile theory, thereby explicitly accounting for correlations among forecast errors.

A. Uncertainty Modeling of Forecast Error

Numerous studies have been undertaken to investigate the modeling of the forecast errors of RESs and loads [20], [28]. Without loss of generality, it is assumed that the probability

distributions of WT, PV, DC load, and RL forecasts are Gaussian, with the mean equal to the forecast value and the standard deviation equal to 30%, 20%, 15%, and 20% of the mean values, respectively. The correlation coefficient between RL and DC load is 0.65. The correlation coefficients between loads and PV output, as well as WT output, are 0.43 and -0.57, respectively. The correlation coefficient between PV output and WT output is assumed to be -0.31.

The forecast errors are modeled by their related probability density functions (PDFs) $f(x)$ and cumulative distribution functions (CDFs) $F(x)$, which can be expressed as:

$$f(x) = (2\pi\sigma^2)^{-0.5} \exp\left[-(x-\mu)^2/(2\sigma^2)\right] \quad (15)$$

$$F(x) = \int_{-\infty}^x f(t)dt \quad (16)$$

where x refers to the uncertain parameter, σ is the standard deviation parameter, and μ is the mean value of the uncertain parameter.

To address the issue of estimation bias in the reserve capacity of DCVPP arising from the correlation, the Gaussian copula is used to accurately deal with the correlation among random variables [29]. Suppose the forecast errors in DCVPP at the same period as an m -dimensional random input vector $\mathbf{X} = [X_1, \dots, X_p, \dots, X_m]^T$ with Spearman correlation matrix $\rho_{\mathbf{X}}$, where X_i represents the i -th random variable.

Firstly, the quasi-Monte Carlo simulation (QMCS) method [30] is employed to generate the sample points $\mathbf{V} = [V_1, \dots, V_p, \dots, V_m]^T$ efficiently from the unified and independent standard normal distributions, where m is the dimension of random input variables, and n is the number of sample points for each input variable.

Then, the standard normally distributed samples $\mathbf{Z} = [Z_1, \dots, Z_p, \dots, Z_m]^T$ with Spearman correlation matrix $\mathbf{L}_Z$ are generated by use of the Cholesky decomposition:

$$\begin{cases} \mathbf{Z} = \mathbf{L}_Z \mathbf{V} \\ 2\sin(\pi\rho_{\mathbf{X}}/6) = \mathbf{L}_Z \mathbf{L}_Z^T \end{cases} \quad (17)$$

Based on the principle of equal probability transformation, convert $\mathbf{Z}$ into the original probability spaces and obtain the correlated samples following different distributions $\mathbf{X} = [X_1, \dots, X_p, \dots, X_m]^T$, using inverse transformation:

$$X_i = F_i^{-1}(\Phi(\mathbf{Z}_i)) \quad (18)$$

where $\Phi(\bullet)$ represents the standard normal CDF, and $F_i^{-1}(\bullet)$ is the inverse function of CDF for the i -th random variable.

B. Risk-based Reserve Scheme for DCVPP

Two-sided quantiles and superquantiles are employed to characterize the tail behavior of forecast errors. For a given confidence level β , the lower and upper quantiles of the random variables are denoted by q_{a1} and q_{a2} , and the corresponding superquantiles, representing the expected deviations in the lower and upper tail regions beyond q_{a1} and q_{a2} , are denoted by sq_{a1} and sq_{a2} , respectively. The mechanism can be explained in Fig. 5. The two-sided quantiles and superquantiles calculation of the random variables are expressed as [31]:

$$\begin{cases} q_{a1} = F^{-1}(\alpha_1) \\ q_{a2} = F^{-1}(1 - \alpha_2) \end{cases} \quad (19)$$

$$\begin{cases} sq_{a1} = (\alpha_1)^{-1} \int_{-\infty}^{q_{a1}} tf(t)dt \\ sq_{a2} = (\alpha_2)^{-1} \int_{q_{a2}}^{+\infty} tf(t)dt \end{cases} \quad (20)$$

$$\alpha_1 + \alpha_2 + \beta = 1 \quad (21)$$

Typically, to facilitate the calculation of q_{a1} , q_{a2} and sq_{a1} , sq_{a2} , QMCS method is employed to sample from the PDF of continuous random variables. This allows for the direct determination of approximate values based on the correlated QMCS samples obtained from the Gaussian copula.

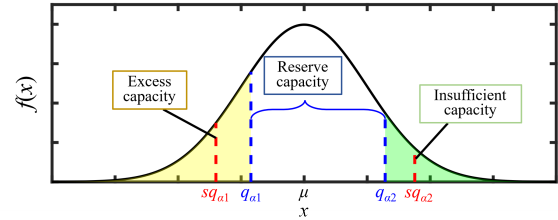


Fig. 5. Mechanism of two-sided quantiles and superquantiles.

For the row vector elements $\mathbf{X}_i = [x_1, x_2, \dots, x_n]^T$ in the generated correlated sample matrix $\mathbf{X}$, sort them in ascending order to obtain the ordered set $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$. The quantiles and the superquantiles can be calculated as:

$$\begin{cases} q_{a1} = x_{(\lfloor \alpha_1 \cdot n \rfloor)} \\ q_{a2} = x_{(\lfloor (1 - \alpha_2) \cdot n \rfloor)} \end{cases} \quad (22)$$

$$\begin{cases} sq_{a1} = (\alpha_1 n)^{-1} \sum_{i=k+1}^n x_{(i)}, \quad k = \lfloor \alpha_1 \cdot n \rfloor \\ sq_{a2} = [(1 - \alpha_2) n]^{-1} \sum_{i=k+1}^n x_{(i)}, \quad k = \lfloor (1 - \alpha_2) \cdot n \rfloor \end{cases} \quad (23)$$

where $\lfloor \cdot \rfloor$ represents the floor function, which rounds a number down to the closest integer. The differences between the two-sided superquantiles and quantiles are used to determine the penalty capacity herein:

$$\begin{cases} P_{\mathfrak{R}}^+(t) = q_{\mathfrak{R}, \alpha_1}(t) - sq_{\mathfrak{R}, \alpha_1}(t), \quad \mathfrak{R} = \{\text{WT, PV, DC, RL}\} \\ P_{\mathfrak{R}}^-(t) = sq_{\mathfrak{R}, \alpha_2}(t) - q_{\mathfrak{R}, \alpha_2}(t), \quad \mathfrak{R} = \{\text{WT, PV, DC, RL}\} \end{cases} \quad (24)$$

where $P_{\mathfrak{R}}^+(t)$ and $P_{\mathfrak{R}}^-(t)$ are the excess capacity and insufficient capacity of the random variable $\mathfrak{R}$, respectively. Note that these reserve capacities are introduced as explicit decision variables, rather than quantities derived from a fixed confidence level.

C. Objective Function

The objective of the proposed optimal scheduling problem is to maximize the net profit of the DCVPP in the electricity-carbon joint market, which is defined as the gross profit Y_{DCVPP} minus the associated costs C_{DCVPP} , and expressed as:

$$\max (Y_{\text{DCVPP}} - C_{\text{DCVPP}}) \quad (25)$$

$$Y_{\text{DCVPP}} = Y^E + Y^{\text{RS}} + Y^{\text{RU}} + Y^{\text{RD}} + Y^{\text{CET}}$$

$$= \sum_{i=1}^{24} \left(\sum_{j \in \{E, \text{RS}, \text{RU}, \text{RD}\}} (p_t^j \cdot P_t^j) + Y_t^{\text{CET}} \right) \quad (26)$$

$$C_{\text{DCVPP}} = \sum_{j \in \{\text{WT}, \text{PV}, \text{ES}, \text{GT}\}} (C_j) + C_L + C_{\text{CO}_2} + C^+ + C^- \quad (27)$$

where Y^E , Y^{RS} , Y^{RU} , Y^{RD} and Y^{CET} are the daily gross profits from the energy, reserve, regulation up, regulation down, and CET market, respectively; p_t^j and P_t^j are the market price and DCVPP's bidding capacity in the market i at time t ; C_j , C_L , C_{CO_2} , C^+ , and C^- are the daily costs of power source j , loads, carbon emissions, excess, and insufficient penalty capacity.

1) Gross Profit of CET Market and Carbon Emission Cost

Under the regulations of CET market, the offset ratio of CCERs to CEAs is capped at 5% of the total allowances being offset. Carbon emissions are initially covered using CEAs. When emissions exceed the allocated CEAs, CCERs may partially cover the excess, while any remaining deficit must be compensated through allowance purchases in the CET market [25]. Furthermore, surplus CEAs and CCERs can be traded to generate additional revenue. Accordingly, the gross profit from CET market participation and the associated carbon emission costs are calculated as follows:

$$\Delta Q_t = Q_t^{\text{CE}} - Q_t^{\text{CEA}} \quad (28)$$

$$C_{\text{CO}_2} = \sum_{i=1}^{24} C_t^{\text{CET}} \quad (29)$$

$$C_t^{\text{CET}} = \begin{cases} 0, & \Delta Q_t \leq r^{\text{CC}} Q_t^{\text{CEA}} \\ \left[Q_t^{\text{CE}} - (1 + r^{\text{CC}}) Q_t^{\text{CEA}} \right] p^{\text{CEA}}, & \Delta Q_t > r^{\text{CC}} Q_t^{\text{CEA}} \end{cases} \quad (30)$$

$$Y_t^{\text{CET}} = \begin{cases} Q_t^{\text{CC}} p^{\text{CC}} - \Delta Q_t p^{\text{CEA}}, & \Delta Q_t \leq 0 \\ (Q_t^{\text{CC}} - \Delta Q_t) p^{\text{CC}}, & 0 < \Delta Q_t \leq r^{\text{CC}} Q_t^{\text{CEA}} \\ (Q_t^{\text{CC}} - r^{\text{CC}} Q_t^{\text{CEA}}) p^{\text{CC}}, & \Delta Q_t > r^{\text{CC}} Q_t^{\text{CEA}} \end{cases} \quad (31)$$

where r^{CC} is the CCER offset ratio, p^{CEA} and p^{CC} are the trading prices of CEA and CCER in the CET market, respectively.

2) Costs of WT, PV, ES, and GT in DCVPP

The generation costs of WT and PV are adopted from [32], while the detailed cost formulations and parameters for ES and GT are available in [18] and [33].

3) Costs of loads (DC and RL) in DCVPP

$$C_L = C_{\text{pur}} + C_{\text{DR}} \quad (32)$$

$$\begin{cases} C_{\text{pur}} = \sum_{i=1}^{24} P_t^{\text{Lpur}} \kappa^E p_t^E \\ C_{\text{DR}} = \delta_\omega \sum_{i=1}^{24} \sum_{t'=t+1}^{24} \omega_{it'} (t' - t) \end{cases} \quad (33)$$

where C_{pur} denotes the cost of power purchased from the retail market; C_{DR} is the cost of the DR program; δ_ω is the cost of migrating per unit batch workload over the time domain; P_t^{Lpur} is the power purchased by the DC load and RL from the retail market. κ^E is the ratio of the retail market price to the energy market price, which is set to 1.45 [18].

4) The Penalty Costs of Excess and Insufficient Power

$$C^+ = \sum_{i=1}^{24} \varphi^+ p_t^E \sum_{\mathfrak{R}} \alpha_{\mathfrak{R}1,t} P_{\mathfrak{R},t}^+ \quad (34)$$

$$C^- = \sum_{i=1}^{24} \varphi^- p_t^E \sum_{\mathfrak{R}} \alpha_{\mathfrak{R}2,t} P_{\mathfrak{R},t}^- \quad (35)$$

where φ^+ and φ^- are the deviation penalty coefficients of excess and insufficient penalty power; $\alpha_{\mathfrak{R}1,t}$ and $\alpha_{\mathfrak{R}2,t}$ denote the probabilities corresponding to the occurrence of the excess and insufficient power scenarios, respectively.

D. Problem Constraints

1) Constraints of DCVPP's Bidding Capacity

$$P_t^E = P_t^{\text{W}} + P_t^{\text{PV}} + P_t^{\text{disch}} + P_t^{\text{GT}} - P_t^{\text{LVPP}} - P_t^{\text{chVPP}} \quad (36)$$

$$P_t^m = P_{\text{ES},t}^m + P_{\text{GT},t}^m, \quad m \in \{\text{RS}, \text{RU}, \text{RD}\} \quad (37)$$

where P_t^{disch} and P_t^{chVPP} represent the discharging and charging capacities of ES; P_t^{LVPP} is the total load power supplied by the sources in DCVPP; $P_{\text{ES},t}^m$ and $P_{\text{GT},t}^m$ are the bidding capacities of ES and GT in the ancillary market m .

2) Constraints of DC Participating in DR Program

Transmission and queuing delays for workloads must adhere to the constraints defined by the M/M/1 queuing theory and not exceed the maximum delay specified in the service level agreement [16]. The related constraints can be described as:

$$0 \leq n_i^{\text{on}} \leq n \quad (38)$$

$$0 \leq u_t \leq u_{\text{max}} \quad (39)$$

$$0 \leq (\mu - E_t^{\text{flex}} / n_i^{\text{on}})^{-1} \leq T_{\text{delay}} \quad (40)$$

where n is the number of servers in DC, u_{max} is the maximum CPU utilization for a single server, and T_{delay} is the tolerant service delay of workloads.

The internal ambient temperatures of the DC are within acceptable ranges, which can be expressed as:

$$\text{Tem}_{\text{min}}^a \leq \text{Tem}_t^a \leq \text{Tem}_{\text{max}}^a \quad (41)$$

$$|\text{Tem}_t^a - \text{Tem}_{t-1}^a| \leq \Delta \text{Tem}_{\text{max}}^a \quad (42)$$

where $\text{Tem}_{\text{min}}^a$ and $\text{Tem}_{\text{max}}^a$ denote the minimum and maximum indoor temperature in the DC, and $\Delta \text{Tem}_{\text{max}}^a$ is the maximum variation of indoor temperature.

Constraints (43) and (44) represent the load-power balance requirements of the DCVPP. Constraint (45) indicates that the DCVPP cannot simultaneously sell electricity in the energy market while its loads are purchasing power.

$$P_t^{\text{DC}} + P_t^{\text{RL}} = P_t^{\text{Lpur}} + P_t^{\text{LVPP}} \quad (43)$$

$$0 \leq P_t^{\text{Lpur}}, P_t^{\text{LVPP}} \quad (44)$$

$$\text{sgn}[P_t^{\text{Lpur}}] + \text{sgn}[P_t^E] \leq 1 \quad (45)$$

where P_t^{DC} and P_t^{RL} are the power demands of DC and RL at time t ; $\text{sgn}[\cdot]$ indicates the signum function.

3) Constraints of Reserve Capacity

The related constraints of the reserve capacity balance of DCVPP can be described as follows.

$$\sum_{\mathfrak{R}} P_{\mathfrak{R},t}^{\text{resup}} = P_{\text{ES},t}^{\text{resup}} + P_{\text{GT},t}^{\text{resup}} \quad (46)$$

$$-\sum_{\mathcal{R}} P_{\mathcal{R},t}^{\text{resdown}} = P_{\text{ES},t}^{\text{resdown}} + P_{\text{GT},t}^{\text{resdown}} \quad (47)$$

$$0 \leq P_{\mathcal{R},t}^{\text{resup}}(t) \leq P_{\mathcal{R}}^{\mathcal{R}} \quad (48)$$

$$P_{\mathcal{R},t}^{\mathcal{R}} - P_{\text{max}}^{\mathcal{R}} \leq P_{\mathcal{R},t}^{\text{resdown}} \leq 0 \quad (49)$$

where $P_{\text{ES},t}^{\text{resup}}$ and $P_{\text{GT},t}^{\text{resup}}$ denote the upward reserve capacity provided by ES and GT; $P_{\text{ES},t}^{\text{resdown}}$ and $P_{\text{GT},t}^{\text{resdown}}$ are the downward reserve capacities provided by ES and GT; $P_{\mathcal{R},t}^{\text{resup}}$ and $P_{\mathcal{R},t}^{\text{resdown}}$ are the upward and downward reserve capacity received by the random power source $\mathcal{R}$; $P_{\text{max}}^{\mathcal{R}}$ is the maximum power of $\mathcal{R}$.

4) Constraints of ES and GT

The constraints and parameters for ES and GT are available in [18] and [34].

V. CASE STUDIES

To validate the effectiveness and superiority of the proposed model and strategy, a series of case studies based on practical data are conducted. The composition of the diversified units within the DCVPP is introduced, followed by the presentation and discussion of the simulation results. In particular, the analysis focuses on several key aspects, including the modeling of the cooling system power consumption in DC, uncertainty modeling using the probability distribution of forecast error, and the optimal scheduling strategy for DCVPP in the electricity-carbon joint market. The problem is coded with the YALMIP toolbox [35] in MATLAB environment and solved by the use of commercial optimization software GUROBI [36].

A. Basic Data

The typical DCVPP shown in Fig. 2 is selected as the test system. The maximum power demands of the DC load and RL are 4 MW and 1.5 MW, respectively, while the installed capacity of GT is 6 MW. The maximum day-ahead forecast power outputs of WT and PV are 4 MW and 4.5 MW, respectively. The maximum power and rated capacity of ES are 6 MW and 15 MWh, respectively. The parameters of the generation units and ES are adopted from [18]. Without loss of generality, it is assumed that 30% of the total arriving workloads are the interactive workloads, 50% are batch workloads, and the remaining 20% are rigid workloads. The detailed parameters of the DC and CET market are presented in Supplementary Material Table S-II and Table S-III, respectively [37] [38]. The WT and PV power generation, load profiles, and ERCOT market prices data for four typical days are shown in Supplementary Material Fig. S-1, where the time step of the sampling point is 1 h [39].

B. Accuracy and DR Capability of DC Load Model

In this subsection, three different DC load models are compared to demonstrate the accuracy and superiority of the proposed DC load model. To ensure the validity of the evaluation results, real operational data collected from a DC in China are used for model accuracy verification. Hourly energy consumption and temperature data are employed as inputs to the second-order ETP model to estimate the corresponding parameters of the DC cooling system power con-

sumption. For comparison, two existing methods are also applied to calculate the cooling system power consumption in DC using the same input data [7], [8]. The first benchmark method adopts a PUE model [7], in which regression analysis is conducted to obtain a static PUE metric representing the ratio of total DC power consumption to IT equipment power consumption. The second benchmark method utilizes a first-order ETP model to characterize the coupling relationship between power consumption of IT equipment and the cooling system [8].

Fig. 6 compares the actual and estimated hourly DC load over 480 hours. The estimation results obtained through the second-order ETP model closely align with the actual data, effectively capturing the dynamic characteristics of DC load behavior. This high level of accuracy demonstrates the capability of the second-order ETP model to characterize the coupled electrical - thermal dynamics of the DC load. Fig. 7 presents the root mean square errors (RMSEs), mean absolute errors (MAEs), and mean absolute percentage errors (MAPEs) of the DC load estimates obtained from the three models. Compared with the two benchmark models, the RMSE of the second-order ETP model is reduced by 68.80% and 57.91% on average, respectively. Similarly, the MAE is reduced by 69.68% and 64.27%, while the MAPE results further confirm the superior estimation accuracy of the second-order ETP model. Collectively, these results validate the superior performance of the proposed model in accurately capturing the dynamic characteristics of the DC load.

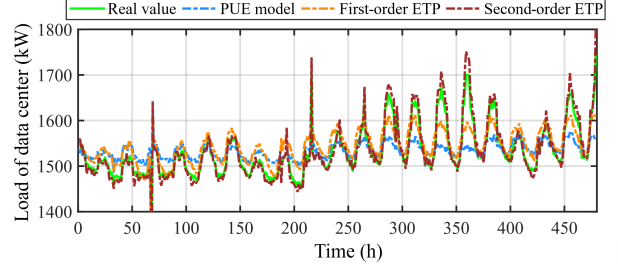


Fig. 6. Practical fitting effects of different DC load models.

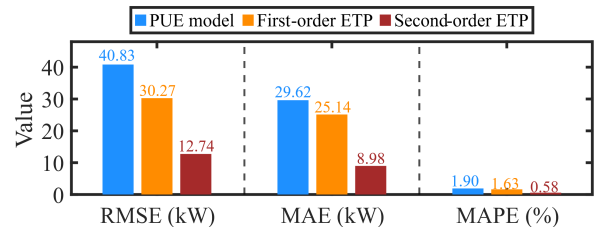


Fig. 7. Comparison of accuracy of DC load models.

To illustrate the practical applicability of the proposed DC load model, three different DC load models (PUE, the first-order ETP, and the second-order ETP models) are incorporated into the optimal scheduling strategy for the DCVPP described in Section IV, corresponding to the scenarios S1-S3, respectively. Fig. 8 compares the DR capability of these scenarios across four seasonally typical days. To provide a benchmark, a scenario without demand response (denoted as Without DR) is also evaluated, where the DC load remains

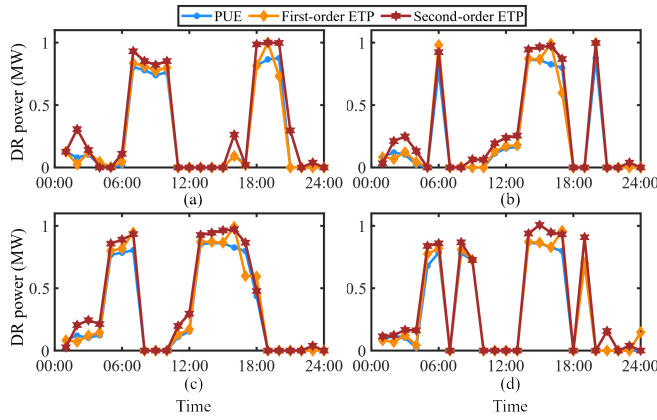


Fig. 8. DR capability comparison of DC load models across typical days in four seasons: (a) spring; (b) summer; (c) autumn; (d) winter.

at its nominal level without any flexibility. As presented in Table I, the total DR capability in S3 exhibits an average increase of 21.6% and 15.9% compared to S1 and S2, respectively. Consequently, the average net profit of DCVPP in S3 is 29.7% and 24.9% higher compared to that of S1 and S2, respectively. These results can be attributed to two primary factors. First, characterizing the relationship between the DC load and the power consumption of IT equipment solely through a static PUE value overlooks its potential for dynamic adjustments in response to actual demand, thereby constraining the flexibility of DC load demand management. Second, the first-order ETP model neglects the effects of temperature variations in the solid materials within the DC, resulting in inaccuracies in representing thermodynamic characteristics and consequently limiting the DR capability of the DC.

TABLE I
OPTIMIZATION RESULTS IN DIFFERENT SCENARIOS.

		Without DR	S1	S2	S3
Spring	DR capability (MW)	0	6.12	6.24	7.74
	Net profit (\$)	439	457	467	541
Summer	DR capability (MW)	0	5.77	6.11	7.15
	Net profit (\$)	97	139	149	216
Autumn	DR capability (MW)	0	7.69	8.07	9.05
	Net profit (\$)	251	354	365	444
Winter	DR capability (MW)	0	7.34	7.82	8.79
	Net profit (\$)	153	182	194	267

C. Analysis of Reserve Capacity in DCVPP

To evaluate the impact of reserve capacity provision on DCVPP operations, a comparative analysis is conducted across three operational scenarios: S4 (without reserve capacity), S5 (with reserve capacity but without considering forecast error correlations), and S6 (with reserve capacity considering forecast error correlations). Considering the typical spring day as an example, Table II provides the optimal day-ahead economic scheduling results in S4-S6.

Comparing net profits with and without reserve capacity, as shown in Table II, indicates that the reserve capacity

TABLE II
IMPACT OF CORRELATION ON OPERATION OF DCVPP IN TYPICAL SPRING DAY.

	Reserve capacity (MW)		Penalty cost (\$)		Net profit (\$)
	excess	insufficient	excess	insufficient	
S4	0	0	645.3	1290.6	-887
S5	17.2	-28.1	53.8	212.3	528
S6	20.1	-32.2	35.6	165.5	541

scheme in S5 and S6 enables the DCVPP to transform losses into profits, thus substantially improving its economic performance. This verifies the effectiveness of the proposed reserve scheme within the optimization strategy. After accounting for forecast error correlations, the total penalty cost in S6 is reduced by 24.41%, and the net profit increases by 2.36% compared with S5. The results across different reserve configurations show that consideration of forecast error correlations improves the estimation of power deviations and reserve adequacy, leading to lower penalty costs and higher net profit across multiple markets. Furthermore, although incorporating correlations produces a reduced penalty cost, the corresponding increase in total net profit is relatively modest. This is due to that neglecting correlations leads to a more aggressive reserve allocation with lower reserve capacity requirements, which increases bidding capacity and consequently enhances revenue in the DAM. As a result, the revenue gains partially offset the penalty cost reductions, yielding only a limited improvement in the expected net profit. However, the offset is severely affected by the deviation penalty coefficients, which are covered in detail in Section V. E. Sensitivity Analysis.

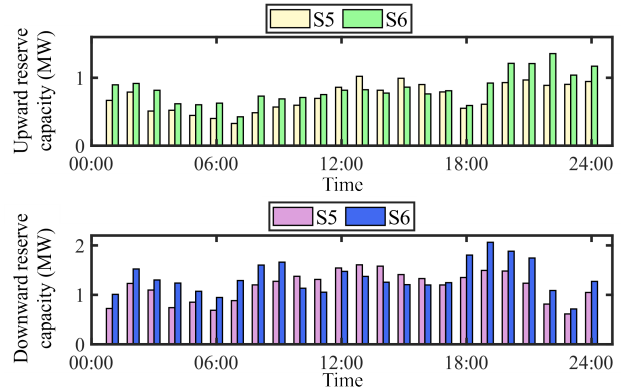


Fig. 9. Comparison of reserve capacity on typical spring day.

Fig. 9 shows the impact of forecast errors correlations on the reserve scheme of the DCVPP for the typical spring day. Due to the strong negative correlation coefficient between WT and load (-0.57), during the period 19:00-6:00, a positive forecast error in WT output (i.e., WT generation being overestimated) is more likely to coincide with a negative forecast error in load demand (i.e., load being underestimated). Consequently, DCVPP in S6 requires a larger upward reserve capacity than in S5 to mitigate the penalty costs associ-

ated with power shortages. Conversely, due to the strong positive correlation coefficient between PV and load (0.43), during the period 10:00-17:00, DCVPP requires a larger downward reserve capacity to reduce the penalty costs arising from power surplus. Notably, the downward reserve capacity significantly exceeds the upward reserve capacity. This is due to that allocating reserve capacity constrains the DCVPP's bidding capacity in the energy, reserve, and regulation up market. Accordingly, the DCVPP optimizes its reserve allocation by balancing multiple markets revenue against penalty costs.

D. Motivational Analysis and Bidding Strategy Optimization for DCVPP Aggregation

To interpret the motivation of such DCVPP aggregation in multiple markets, this study designs distinct test scenarios in four typical days as mentioned in Table III, thus systematically evaluating the impact of DC load involved in coordination with the VPP aggregation and DC's participation in DR programs.

It is depicted in Table III that the negative net profit for the case where the DC is not aggregated with VPP arises from the high cost of DC purchasing electricity from the energy market. However, when the VPP and DC are integrated into a DCVPP, the DC load can be supplied by the RESs within the VPP. This integration reduces the energy cost of DCVPP and increases the net profit to \$541, \$216, \$444, and \$267 in the four seasonal days, respectively. Thus, aggregating DC into the VPP yields substantial economic benefits. Moreover, the DC participation in the DR program has a significant impact on the net profit of DCVPP. Comparative results show an average profit increase of 74.4% with DR implementation compared to the scenario without DR, primarily due to the strategic temporal shifting of batch workloads through time-domain migration.

TABLE III

ECONOMIC COMPARISON OF DIFFERENT COMBINATIONS OF DCVPP IN FOUR TYPICAL DAYS.

	Net profit (\$)	DC and VPP (Without aggregation)	DCVPP (With aggregation)
Spring	Without DR	-835	439
	With DR	-485	541
Summer	Without DR	-973	97
	With DR	-604	216
Autumn	Without DR	-819	251
	With DR	-400	444
Winter	Without DR	-927	153
	With DR	-562	267

The comparisons of the workload distribution patterns and load power of DCVPP before and after DR in four typical days are presented in Supplementary Material Fig. S-2 and Fig. S-3, respectively. Considering the typical spring day as an example, during the periods of peak electricity prices (07:00-10:00 and 17:00-19:00), the DCVPP strategically reallocates workloads to the low-price periods (12:00-15:00 and

21:00-24:00). This load-shifting mechanism serves dual purposes: it reduces the operating costs while enhancing DCVPP's bidding capacity in the DAM.

To validate the effectiveness of proposed optimal scheduling strategy for DCVPP participating in the electricity-carbon joint market, three comparative cases are designed in Table IV. The corresponding optimization results under different market portfolio cases are presented in Table V.

By comparing the net profits of Case 1 and Case 2 in Table V, it can be observed that when the DCVPP participates only in the energy market without engaging in reserve and regulation up/down market, its flexible regulation capability is not fully utilized, resulting in overall losses. In contrast, coordinated participation in the diversified DAM increases DCVPP's profit from \$-471, \$-750, \$-355, and \$-944 to \$279, \$-11, \$192, and \$-31, respectively, thus highlighting the significance of its participation in diversified electricity markets. In addition, Table V illustrates the impact of the CET mechanism on the DCVPP's profit and carbon emissions. Compared with Case 2, the daily carbon emissions in Case 3 are reduced by 5.6%, 7.6%, 1.2% and 41% in four typical days, while the net profit increases by \$261, \$226, \$252, and \$298, respectively. The improvements can be attributed to the synergistic effects of the electricity-carbon trading mechanism, whereby the DCVPP strategically reduces carbon emission costs through decreased GT utilization while obtaining additional carbon asset income via the RES integration for CCER trading. Consequently, the incorporation of the CET mechanism effectively aligns the economic and environmental objectives of the DCVPP.

TABLE IV
DESIGN OF DIFFERENT MARKET PORTFOLIO CASES.

Case	DAM			CET market	
	Energy	Reserve	Regulationup/down	CEA	CCER
1	√				
2	√	√	√		
3	√	√	√	√	√

"√" represents DCVPP's participation in the corresponding market.

Across the four seasonally typical days, the spring day exhibits the highest overall revenue due to peak RES outputs, which maximizes its bidding capacity and profit through the diversified DAM. Compared with the spring day, the autumn day demonstrates marginally lower RES output and DAM prices, resulting in slightly reduced revenue. In summer, heavy reliance on PV output creates significant diurnal volatility combined with peak power loads, necessitating frequent GT unit activation during RES output valleys, thereby elevating its operational costs and restricting overall profitability. In winter, constrained RES generation capacity limits DAM bidding participation and profit optimization shifts to maximizing CET gains through strategic carbon emission reduction.

Fig. 10 presents the detailed bidding capacity allocations of the DCVPP in various electricity markets across four seasonally typical days. Due to variations in renewable energy

TABLE V
ECONOMIC AND EMISSION COMPARISON UNDER DIFFERENT MARKET COMBINATIONS FOR FOUR TYPICAL DAYS.

	Case	DAM revenue (\$)	CET revenue (\$)	Operating cost (\$)	Carbon emissions(tCO ₂)	Net profit(\$)
Spring	1	4147	\	4618	114	-471
	2	4599	\	4320	93	279
	3	4472	269	4200	88	541
Summer	1	3336	\	4086	96	-750
	2	4028	\	4039	84	-11
	3	3866	240	3890	78	216
Autumn	1	3281	\	3636	80	-355
	2	3703	\	3511	68	192
	3	3612	265	3433	67	444
Winter	1	3055	\	3999	94	-944
	2	3796	\	3827	76	-31
	3	2926	350	2909	44	267

availability and load demand, the bidding strategies differ markedly on these days. It is evident that the bidding strategy in the DAM dynamically matches the peak and valley time of the market price curves, as reflected in Supplementary Material Fig. S-1. When market prices are relatively high (05:00-13:00 in spring and 14:00-16:00 in summer), the DCVPP prefers to sell more electricity correspondingly. Conversely, during peak load periods (1:00-4:00 in both autumn and winter), the DCVPP reduces its bidding capacity to avoid internal power shortage. In addition, when ancillary service market prices exceed those of the energy market (05:00 - 07:00 in winter), the DCVPP strategically reallocates the bidding capacity toward ancillary service provision. As further evidenced in Supplementary Material Fig. S-3, the DC workloads are shifted from the peak-price periods to the low-price periods, thereby enhancing DCVPP's bidding capacity during peak-price periods. These results indicate that the proposed strategy effectively exploits the potential flexibility of DCVPP and enables a coordinated optimization across multiple markets, ultimately improving operational economy performance.

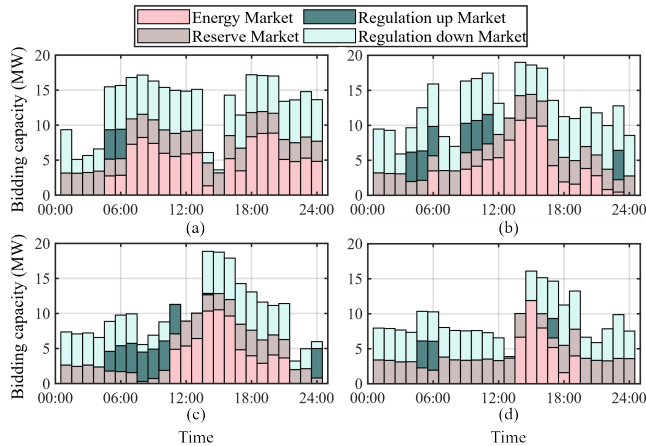


Fig. 10. Bidding capacity of DCVPP in DAM across typical days in four seasons. (a) Spring. (b) Summer. (c) Autumn. (d) Winter.

E. Sensitivity Analysis

To ensure both conciseness and representativeness, the sensitivity analysis is conducted on the spring typical day, which reflects moderate conditions of RES generation and DAM prices. The analysis quantifies the impacts of key parameters on operation results to evaluate the robustness of the proposed model and strategy.

To investigate the influence of the deviation penalty coefficient on the reserve scheme and daily net profit of DCVPP, a sensitivity analysis is conducted under different multiples of the penalty coefficients, which are designed according to the originally preset penalty coefficients of $\phi^+=1.0$, $\phi^-=2.0$ [18].

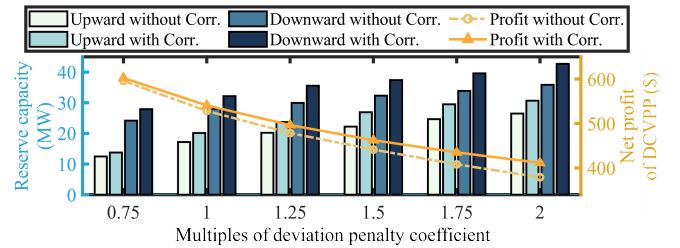


Fig. 11. Impact of deviation penalty coefficient on reserve scheme and profit with and without considering correlation on the typical spring day.

As shown in Fig. 11, both the upward and downward reserve capacities consistently increase with rising deviation penalty coefficients on the typical spring day. This indicates that as the DCVPP operator becomes more risk-averse, larger reserve capacities are required to hedge against uncertainty. Meanwhile, the net profit of DCVPP decreases since higher penalty coefficients effectively increase the weight of the penalty cost term in the objective function. Notably, with increasing the deviation penalty coefficients, the total reserve capacity considering the correlation increases more significantly, and the profit reduces less compared to the correlation-ignored method, thus verifying the applicability of the proposed reserve scheme under different risk preferences.

To assess the impact of adjusting the CCER offset ratio

on DCVPP revenue, the original baseline ratio (5%) was increased and decreased by 20% and 40%, corresponding to offset ratios of 9%, 7%, 3%, and 1%, respectively. The resulting DCVPP profit and carbon emissions after implementing these offset ratio adjustments are presented in Supplementary Material Fig. S-5. When the CCER offset ratio increases from 1% to 9%, the net profit of DCVPP demonstrates an upward trend. The minimum profit of \$532.18 occurs at a 1% offset ratio, while the maximum profit of \$549.46 is achieved at a 9% offset ratio, representing a variation of 3.25%. Carbon emissions exhibit an increasing trend with higher CCER offset ratios, reaching a minimum of 87.67 t at 1% and a maximum of 87.72 t at 9%. This phenomenon occurs because higher CCER offset ratios enable the DCVPP to purchase more low-cost CCER credits for compliance, thereby reducing its carbon emission costs and increasing profits. However, paradoxically, excessive reliance on the CCER offset mechanism may diminish the DCVPP's incentive to implement emission reduction measures, potentially undermining decarbonization efforts.

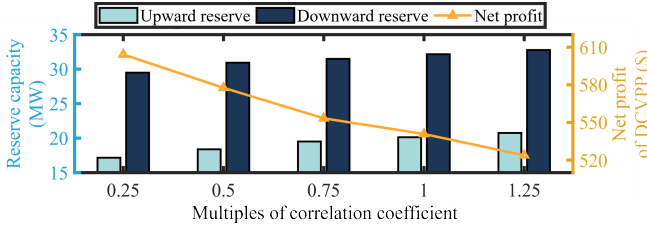


Fig. 12. Sensitivity analysis of reserve requirements and net profit under different correlation levels on typical spring day.

To quantify the impact of correlation among RESs and loads forecast errors, the baseline correlation matrix is scaled by 0.25, 0.50, 0.75, 1.00, and 1.25, while keeping all marginal distributions and model settings unchanged. All scaled matrices remain positive semidefinite. For each scenario, the uncertainty model is regenerated and the DCVPP scheduling problem is solved. As shown in Fig. 12, higher correlation levels consistently result in larger upward and downward reserve requirements, since the stronger correlation increases the likelihood of simultaneous forecast deviations, requiring more reserve capacity to hedge the joint risks. Consequently, less flexible capacity is available for market trading, leading to a decline in DCVPP's net profit as the correlation strength increases. These results indicate that the proposed model and strategy respond appropriately to changes in correlation structure and maintain robust performance across the considered correlation range.

VI. CONCLUSION

This paper presents an operational framework for DCVPP that integrates DC with a VPP consisting of WT, PV, ES, GT, and RL. Furthermore, an optimal scheduling strategy is proposed for the DCVPP participating in the electricity-carbon joint market under uncertainty. From the theoretical models and case study results, the main conclusions are drawn as follows:

1) Compared with two existing DC load models, the proposed model achieves a substantially improved fitting accuracy, with the RMSE decreased by 68.80% and 57.91%, and the MAE by 69.68 % and 64.27%. For the cases adopting the proposed model, the net profit of DCVPP increases by an average of 36.5% and 30% across four seasonally typical days. The results illustrate that the proposed model based on the second-order ETP theory can more accurately capture the load characteristics of DC and effectively exploit its DR potential.

2) Based on the proposed risk-based reserve scheme incorporating forecast error correlations, the penalty cost of the DCVPP is reduced by 24.41%, while the net profit increases by 2.36% compared to the scheme that neglects such correlation. Test results verify that the proposed scheme can mitigate the penalty risk of excess and insufficient power caused by forecast errors of RESs and load, thereby enhancing the economic performance of the DCVPP across multiple markets.

3) The proposed optimal scheduling strategy for such aggregated DCVPP significantly increases overall profitability while reducing carbon emissions in the electricity-carbon joint market. Test results illustrate that the net profit of DCVPP is increased by 74.4% on average when the DR program is enabled. Moreover, coordinated participation in the diversified DAM increases DCVPP's net profit by \$261, \$226, \$252, and \$298 for four typical days. In addition, the integration of the CET mechanism aligns the economic benefits with low-carbon objectives in the optimal scheduling of the DCVPP.

Overall, the results indicate that the proposed strategy is well-suited for practical DCVPP operations in the joint electricity – carbon markets. For DC operators, it systematically exploits workload flexibility to provide DR while maintaining service continuity and operational reliability. For VPP aggregators, it enables coordinated management of diverse resources, thus enhancing revenue robustness and reducing imbalance risks under forecast uncertainty. For policymakers and market designers, the framework provides an implementable approach to integrating carbon trading with DR, supporting policy designs that jointly pursue economic efficiency and verifiable carbon emission reductions.

Future work will explore the extension to DCVPPs based on multiple DCs, enabling such spatial flexibility through the coordinated DR and workload migration across geographically distributed DCs. In addition, incorporating grid constraints and real-time carbon markets will further enhance the practical applicability and decarbonization effectiveness.

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